

Member ID: _____

Time: _____

Rank: _____



FUNDAMENTALS OF WEB DESIGN (415)

STATE 2024

CONCEPT KNOWLEDGE:

Multiple Choice (25 @ 2 points each) _____ (50 points)

APPLICATION KNOWLEDGE:

Application _____ (250 points)

TOTAL POINTS _____ (***300 points***)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. Which tag is used in HTML5 to play video files?
 - A. <video>
 - B. <media>
 - C. <mp4>
 - D. <film>
2. In CSS3, how do you select an element with the id "banner"?
 - A. banner {}
 - B. .banner {}
 - C. #banner {}
 - D. *banner {}
3. Which of the following is NOT a valid HTML5 tag?
 - A. <section>
 - B. <header>
 - C. <blink>
 - D. <article>
4. What is the purpose of the 'alt' attribute in an image tag in HTML?
 - A. To create a border around the image.
 - B. To specify an alternate text for the image if it cannot be displayed.
 - C. To change the image source.
 - D. To add a caption below the image.
5. In CSS3, which property is used to change the text color of an element?
 - A. text-color
 - B. color
 - C. font-color
 - D. background-color
6. Which HTML5 tag is used to specify a footer for a document or section?
 - A. <footer>
 - B. <bottom>
 - C. <foot>
 - D. <end>
7. What does the 'href' attribute in an anchor tag specify?
 - A. The type of hyperlink
 - B. The destination URL of the hyperlink
 - C. The color of the hyperlink
 - D. The name of the hyperlink

8. In CSS3, which property is used to make the corners of an element rounded?
- A. border-radius
 - B. round-corners
 - C. corner-curve
 - D. edge-round
9. Which attribute is used in HTML to specify that an input field must be filled out before submitting a form?
- A. required
 - B. mandatory
 - C. validate
 - D. necessary
10. In CSS3, how do you apply a style for all <h1> elements of class 'main'?
- A. h1.main {}
 - B. h1 + .main {}
 - C. .main h1 {}
 - D. main.h1 {}
11. What is the correct HTML5 element for playing audio files?
- A. <mp3>
 - B. <sound>
 - C. <audio>
 - D. <music>
12. Which CSS property is used to control the opacity of an element?
- A. opacity
 - B. filter
 - C. visibility
 - D. transparency
13. In HTML, which attribute is used to specify that an input field is read-only?
- A. readonly
 - B. fixed
 - C. immutable
 - D. static
14. Which of the following is a self-closing tag in HTML5?
- A.
 - B. <div>
 - C. <p>
 - D. <h1>

15. In CSS3, what is the correct syntax to select all <p> elements inside <div> elements?
- A. div + p {}
 - B. div p {}
 - C. p > div {}
 - D. div.p {}
16. What is the purpose of the 'lang' attribute in an HTML document?
- A. To define the character encoding.
 - B. To specify the language of the document's content.
 - C. To set the language for spell check.
 - D. To define the language for the browser's interface.
17. Which HTML5 element is used for marking up a navigational link area?
- A. <navigation>
 - B. <navigate>
 - C. <nav>
 - D. <links>
18. In CSS3, which property is used to add shadow to text?
- A. text-shadow
 - B. font-shadow
 - C. shadow
 - D. text-style
19. Which attribute is necessary to make an HTML5 <canvas> element functional?
- A. context
 - B. width and height
 - C. style
 - D. controls
20. In CSS3, which property is used to transform elements in 2D or 3D?
- A. transform
 - B. translate
 - C. transition
 - D. alter
21. Which of the following is NOT a valid attribute for the <table> tag in HTML?
- A. border
 - B. cellpadding
 - C. bgcolor
 - D. align

22. In HTML, which tag is used to create a numbered list?
- A.
 - B.
 - C.
 - D. <dl>
23. Which CSS property is used to control the space between lines of text?
- A. text-indent
 - B. line-height
 - C. letter-spacing
 - D. word-spacing
24. In HTML5, which attribute is used with the <meta> tag to define the character set of the document?
- A. charset
 - B. content
 - C. http-equiv
 - D. name
25. Which CSS property is used to create a transition effect when changing from one style to another?
- A. transition
 - B. transform
 - C. animation
 - D. change

APPLICATION KNOWLEDGE

Overview

You are tasked with creating a modern, responsive web page titled "Explore Chicago." This page should showcase various attractions, historical sites, and fun facts about Chicago, using advanced HTML5 and CSS3 techniques to create a visually appealing, accessible, and informative experience.

Design Requirements

1. Modern Layout:
 - Employ Flexbox for layout design in various sections.
 - Include a **header**, **main**, and **footer**.
2. Navigation Bar:
 - Located in the **header**.
 - Create jump links for the following: **Attractions**, **History**, **Fun Facts**, **Contact**
3. Showcase Area:
 - A banner section with a background image ("images/chicago-banner.jpg").
 - Heading: "Welcome to Chicago" and subheading: "Explore the city of winds".
4. Content Sections:
 - Located in the **main**.
 - Three sections in **main**: **Attractions**, **History**, **Fun Facts**
 - **Attractions**
 - a. Title
 - i. Discover the Wonders of Chicago
 - b. Text Content
 - i. Willis Tower: An iconic skyscraper offering stunning skyline views. A must-visit for breathtaking sights.
 - ii. The Art Institute of Chicago: Home to an impressive collection of artworks, spanning thousands of years.
 - iii. Navy Pier: A vibrant location filled with amusement rides, shops, and eateries, perfect for family fun.
 - c. Image
 - i. chicago-attractions.jpg
 - d. Table Content
 - i. Landmark, Opening Hours, Ticket Price
 - ii. Willis Tower, 9 AM - 10 PM, \$25
 - iii. The Art Institute of Chicago, 10 AM - 5 PM, \$20
 - iv. Navy Pier, 10 AM - 8 PM, Free Entry
 - **History**
 - a. Title
 - i. A Journey Through Time in Chicago
 - b. Text Content
 - i. From its founding in 1833 to becoming a bustling metropolis, Chicago's history is as rich as it is diverse.

- ii. Great Chicago Fire (1871): A major event that led to the extensive rebuilding of the city.
 - iii. World's Columbian Exposition (1893): A landmark world fair that showcased Chicago's resurgence.
 - c. Image
 - i. chicago-history.jpg
 - **Fun Facts**
 - a. Title
 - i. Fun Facts About Chicago
 - b. Text Content
 - i. Chicago's nickname "The Windy City" was coined not just for its weather, but also for its long-winded politicians.
 - ii. The first Ferris wheel made its debut in Chicago in 1893.
 - iii. Chicago River is the only river in the world that flows backward, a result of an engineering marvel.
 - c. Image
 - i. chicago-funfacts.jpg
- 5. Styling:
 - External stylesheet.
 - Consistent color scheme and typography.
- 6. Images:
 - All images must have alt text with a basic photo description.
 - Images should be stored and called from the **images** folder.
- 7. Footer:
 - Contact information
 - a. Contact Information: Chicago Tourism Office, 123 City St, Chicago, IL, 60601, (312) 555-1234, info@chicagotourism.com
- 8. Accessibility and Responsive Design:
 - Ensure website accessibility (text alternatives for images, keyboard navigability).
 - Use media queries for responsiveness on different devices.

Criteria	Description	Points Possible	Points Earned
Layout and Structure	Modern layout using Flexbox, correct implementation of header, main, footer.	40	
Responsiveness	Effective use of media queries, consistent design on various screen sizes.	30	
Navigation Bar	Correct placement and linking, CSS transitions for hover effects.	20	
Showcase Area	Inclusion and styling of banner with image and text, creativity, and visual appeal.	30	
Content Sections	Accurate replication of provided content, styling, and layout of sections.	50	
Styling	Use of modern design elements, consistency in color scheme and typography.	30	
Images and Tables	Correct implementation and alt text for images, effective use of a table in Attractions section with specific landmarks, hours, and pricing.	25	
Footer	Inclusion of contact information.	15	
Accessibility	Adherence to accessibility standards.	20	
	Total	250	